

Efficient Rendering of Glossy Materials by Interpolating Prefiltered Environment Maps based on Primary Normals

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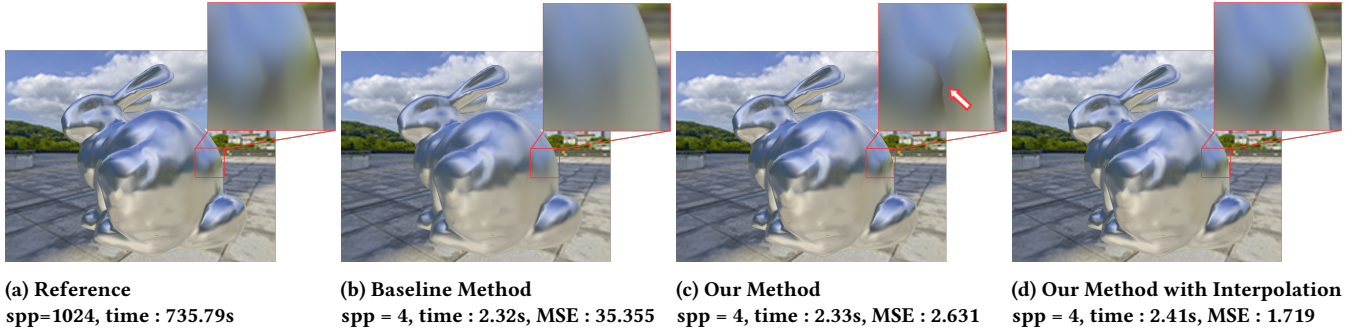


Figure 1: Rendering results of reference and each method. Compared with a reference image (a) rendered by Mitsuba2[Nimier-David et al. 2019], the baseline method (b) which is a naïve prefiltering method improves the speed of rendering however, artifacts are noticeable (see the red boxes). Although, our method of multiple prefiltered environment mapping (c) reduces the mean square error (MSE), still aliasing artifact (see red arrow) remains. By interpolating of above maps, we archive more plausible.

ABSTRACT

We propose a method to improve the speed of the rendering of glossy surfaces including anisotropy reflection, using prefiltering. The key idea is to prefilter an environment light map for multiple primary normals. Furthermore, we propose an interpolation method to smoothly connect the boundaries generated by switching multiple prefiltered environment maps. As a result, we are able to render objects with glossy surface more accurately even in real-time.

CCS CONCEPTS

• **Computing methodologies** → *Real-time simulation; Visual analytics.*

KEYWORDS

digital twin, glossy surface, rendering, prefiltering

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1 INTRODUCTION

Digital twinning has been attracting attention for its efficiency in design process of commercial products. However, the glossy appearance of metallic objects is difficult to visualize due to its complicated reflection especially in real-time. Heidrich[Heidrich and Seidel 1999] approximated the Phong model, while Kautz and McCool[Kautz and McCool 2000] approximated the more general glossy reflections by prefiltering environment light maps to speed up the rendering. However, their methods have large errors due to distortion and can only apply to isotropic materials. In this paper, we propose a method to efficient render glossy materials. By interpolating prefiltered maps based on uniformly selected primary normal directions, it is possible to render the materials more accurately.

2 METHOD

2.1 Prefiltering for each primary normal

Kautz and MacCool [Kautz and McCool 2000] proposed an approximated rendering method to reduce the rendering cost by using prefiltered omnidirectional environment light map which filter kernel depends on the normal distribution of materials to be rendered. However, it is difficult to avoid the rendering artifacts by using a single environment map due to the distortion. To solve this problem, we prepare multiple prefiltered environment maps based on selected multiple primary normal directions.

The illuminations from the directions on hemisphere against the primary normal direction are not generally sampled for rendering process. Therefore, it is possible to combine two prefiltered maps generated by two oppositely oriented primary normal directions, and thus, we can save the memory usage more efficiently (Figure 2). According to our rendering equation, the radiance to the camera direction $L_r(\omega_o)$ can be described:

$$L_r(\omega_o) = \int_H L'_i(\omega_i) \frac{\delta(\omega_i - \omega_{or})}{\omega_i \cdot \mathbf{n}} \omega_i \cdot \mathbf{n} d\omega_i \quad (1)$$

$$= L'_i(\omega_{or}) \quad (2)$$

$$L'_i(\omega_i) = \frac{\int_H L_i(\omega_{ik}) D(\mathbf{h}) d\omega_{ik}}{\int_H D(\mathbf{h}) d\omega_{ik}} \quad (3)$$

where L'_i is the prefiltered environment map and ω_{or} is the mirror reflection vector of ω_o . $D(\mathbf{h})$ is the normal distribution function, and the Beckmann distribution is used in this experiment.

2.2 Interpolation of environment maps

In the rendering process, once a single environment map is selected according to the normal direction, aliasing artifact tends to be appeared due to the selection uncertainty around the boundary. To reduce the artifact, we introduce an interpolation method by weighted summation of multiple environment maps:

$$L_{in} = \sum_k L_{in_k} (\mathbf{n} \cdot \mathbf{n}_k)^a \quad (4)$$

where L_{in_k} is the prefiltered environment map corresponding to the primary normal $\mathbf{n}_k$. Note, L_{in} needs to be normalized before the rendering. The parameter a controls the weight of the interpolation.

3 EXPERIMENT

Rendering experiments were conducted to verify the effectiveness of our method. In the experiment, we selected uniformly sampled 42 primary normals, then generated corresponding environment maps (Figure 2). Note, the number of environment maps can be reduced to half as stated in previous section. Also, it requires additional one environment map without any filterings which is used in case of a large incident angle to a normal. As a result, we used 22 environment maps for this experiment. Figure 1 (a) shows the reference image of the glossy material Stanford Bunny rendered by Mitsuba2 [Nimier-David et al. 2019]. The rendering speed is significantly improved by the naïve prefiltering method in (b), however, our method (c) improves the rendering accuracy. Thanks to the interpolation, we moreover improve noticeable artifacts due to the boundary of multiple environment maps shown in (d).

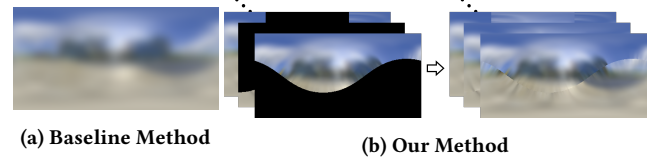


Figure 2: Series of images (b)-left are the prefiltered environment maps of our method generated by each corresponding primary normal directions. Note, half of the regions of each maps are not used for the rendering, thus we can combine two maps into one (b)-right to reduce the memory usage.

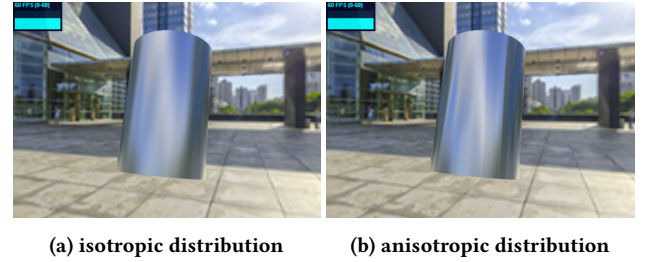


Figure 3: Implementation in WebGL and Anisotropy support. We implemented our method in WebGL. (a) and (b) are rendering results for metal surfaces with isotropic and anisotropic normal distribution. You can see that it can render in real-time and supports anisotropic glossy surface (b).

We also implement our technique in GLSL on WebGL as shown in Fig. 3. The frame rate of our method is over 60fps under the Intel Iris Xe Graphics. This figure also shows that our method can handle objects with anisotropic glossy material which is also an advantage of our work.

4 SUMMARY AND FUTURE WORKS

In this study, we proposed a method to efficiently render glossy reflections by using interpolated multiple prefiltered maps based on primary normals. The experimental results demonstrated the efficiency of the rendering speed. Also, our method can reduce the memory usage. Our method is also possible to handle anisotropic normal distributions which is also the advantage of our work.

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